

3.7 PHYLUM-ARTHROPODA

- *Tagmatization of the body; jointed legs.*

3.7.1 General Characters

This is the **largest phylum** of the Kingdom: Animalia, and it includes the **largest Class** called **Insecta**. Over two-thirds of all named species on earth are arthropods accounting for 80% of the animal species (Figure 3.19 to 3.25). They are bilaterally symmetrical, triploblastic, metamerically segmented and coelomate (**schizocoelomate**) animals. The body of arthropods is covered by a chitinous exoskeleton as a protection and to prevent loss of water and it is periodically shed off by a process called **moulting** or **ecdysis** to allow growth of the body. The body is segmented and consists of *head*, *thorax* and *abdomen*. They have jointed appendages (arthros : joint; podium : foot). Muscles are striated and aid in rapid locomotion. (**Note: striated muscles appeared for the first time in evolution, in the arthropods**). Body cavity is a **haemocoel**. It is a cavity filled with haemolymph. It is formed by the fusion of embryonic blastocoel with some coelomic spaces. The coelom is reduced to the spaces around excretory organs and gonads. Respiratory organs are *gills*, *book gills*, *book lungs* or *tracheae*. Circulatory system is of **open type**. Heart is dorsal in position. In some (crustaceans and chelecerates) the haemolymph contains a ‘*copper*’ containing respiratory pigment called ‘**haemocyanin**’ dissolved in the haemolymph. Nervous system is of annelidan type consisting of a *nerve ring* (around oesophagus) and a double ventral nerve cord. Sensory organs like antennae, eyes (compound and simple), statocysts (organs of balance/ equilibrium) are present. Excretion takes place through **malpighian tubules**, **green glands**, **coxal glands**, etc. They are mostly dioecious (unisexual / gonochoric). Fertilisation is usually internal. They are mostly oviparous. Development may be direct or indirect. Life history includes one to many larval stages followed by metamorphosis. Examples: *Periplaneta*, *Palaemon*, *Cancer*, *Palamnaeus*, *Aranea*; **Economically important insects** – *Apis*, *Bombyx*, *Kerria* (*Laccifer*); **Vectors of common pathogens** – *Anopheles*, *Culex* and *Aedes*; **Gregarious pest** – *Locusta*; **Living fossil** – *Limulus*.



3.7.2 Classification of the phylum Arthropoda

Phylum **Arthropoda** is classified into **three sub-phyla**, **1. Trilobita** **2. Chelicerata** and **3. Mandibulata**.

Sub-phylum-1: Trilobita

They are extinct marine arthropods. Appendages are numerous, similar and *biramous*. Body is divided into a median and two lateral lobes by two prominent *longitudinal furrows*, hence the name trilobite.

Examples: *Triarthrus*, *Dalmanites*.



Fig. 3.19 Trilobite (fossil)

Sub-phylum-2: Chelicerata

Body consists of an anterior six segmented **cephalothorax** or **prosoma**, a posterior thirteen segmented **abdomen** or **opisthosoma** which is further divided into **mesosoma** and **metasoma**. Prosoma bears six pairs of appendages; the first pair is modified for feeding (**chelicerae**). Antennae and jaws (mandibles) are absent. Respiratory organs are **book-gills**, **book-lungs** and **tracheae**.

The sub-phylum **Chelicerata** is classified into **two classes**, namely **1. Xiphosura** and **2. Arachnida**.

Class-i: Xiphosura

They are marine organisms with branchial respiration; includes both **extinct** and **only three genera of extant** (living) animals. Prosoma is covered by broad horse-shoe shaped **carapace**; bears a pair of chelicerae and four pairs of walking legs and one pair

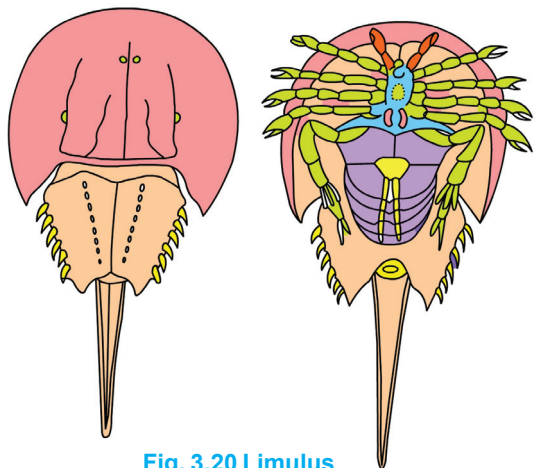


Fig. 3.20 Limulus

of '**pusher legs**'. Mesosoma bears six pairs of appendages, the first pair is fused to form **genital operculum** and the other five pairs are modified into **book-gills**. Metasoma ends in **telson**. Development includes **trilobite larva** in *Limulus*.

Examples: *Limulus* (king crab, a living fossil).

Class-ii: Arachnida

They are terrestrial; prosoma bears a pair of chelicerae, a pair of pedipalpi and four pairs of walking legs. Mesosomal appendages are modified into **book-lungs**. Four pairs of posterior abdominal appendages are modified into **spinnerets** in spiders. Respiratory organs are **book-lungs** (scorpions and some spiders), **tracheae** (some spiders) or **both** (some spiders). Respiratory pigment is '**copper**' containing **haemocyanin**. Excretory organs are **Malpighian tubules** and **coxal glands**. Development is **direct**; scorpions are **viviparous**.

Example: *Palamnaeus* (scorpion), *Aranea* (spider), *Sarcoptes* (itch mite).



Fig. 3.21 Scorpion

Sub-phylum-3: Mandibulata

They are aquatic (freshwater or marine) or terrestrial. Body is divided into two (cephalothorax and abdomen) or three (head, thorax and abdomen) parts. Head bears antennae, mandibles and maxillae. Respiratory organs are gills or tracheae.

The sub-phylum Mandibulata is classified into four classes **1. Crustacea** **2. Chilopoda** **3. Diplopoda** and **4. Insecta**.

Class-i: Crustacea

They are aquatic. Head and thorax fuse forming the **cephalothorax** (covered by chitinous **carapace**). In some the exoskeleton is hardened by calcium carbonate (crabs and lobsters). Cephalic region bears two pairs of antennae (**antennules** and **antennae** – **a unique feature**), one pair of mandibles and two pairs of maxillae. Thoracic and abdominal appendages are '**biramous**'. Respiratory organs are gills (branchiae). Excretory organs are **green glands** or **antennary glands**. Sense organs include **antennae**, **compound eyes**, **statocysts**, etc. Development is **indirect** and includes different larval forms.

Examples: *Palaemon* (freshwater prawn), *Cancer* (crab), *Balanus* (rock barnacle), *Sacculina* (root-headed barnacle), *Astacus* (cray fish), *Daphnia* (water flea).



Fig. 3.22 Prawn

Class-ii: Chilopoda

Chilopoda includes **centipedes** (*hundred leggers*). They are terrestrial, air-breathing, carnivorous animals. Body consists of head and trunk. Each segment of the trunk bears a pair of clawed appendages; first pair of trunk appendages bears **poison claws**. Respiratory structures are *tracheae*. Excretory organs are **Malpighian tubules**.

Examples: *Scolopendra*, *Scutigera*.



Fig. 3.23 Scolopendra

Class-iii: Diplopoda

Diplopoda includes **millipedes** (*thousand leggers*). They are terrestrial, air-breathing animals, feeding on decaying plant material. Body consists of head and trunk. Head bears paired antennae, mandibles and maxillae (maxillae are modified into **gnathochilarium**). Each trunk segment bears two pairs of legs. Respiratory organs are *tracheae*. Excretory organs are **Malpighian tubules**.

Examples: *Spirostreptus*, *Julus*.



Fig. 3.24 Spirostreptus